DSO183 â stock v6 vs custom firmware

Read-only audit, 2026-11-04. Evidence: v6 flash backup, live register reads on the stock image, current ELF and logs. Nothing was built, flashed or modified for this audit.

1. Capability comparison

Area	Capability	Stock v6	Ours	Notes
Acquisition	12-bit ADC on PA2	â	Present (verified SQR3=2)	â	Present (same channel)
Hardware-paced sampling	Present	â	TIM2_CC2 (CR2=0x0098F101, EXTSEL=4)	â	Present	â	TIM3 TRGO (verified)	â	Different timers, same approach
Circular DMA capture	Present	â	Present, 1024-sample ring
ADC sample time	1.5 cycles (SMPR2=0)	â	7.5 cycles signal, per-range policy	â	Ours trades speed for settling
Analog-watchdog trigger	Present (SCAN=1, AWDEN, AWDCH=2)	â	Present, band ´–4 counts (to 1)
Trigger modes AUTO/NORM/SINGLE	Probable (GUI)	â	Partial	â	FSM tested, not wired
Timebase	Range	â	1 ´µs/div â† 500 s/div (published)	â	Same 27-step table (verified)
Selection	GUI (user statement)	â	Software only (SET_TIMEBASE)	â	Unfinished on ours
Display	Controller	â	0x858552, MADCTL 0x68	â	Same (verified)	â	Not an ILI9341
Grid / trace / marker	Unverified detail	â	10ˆ 8 grid, column-mean trace, dashed marker
On-screen numeric readouts	Probable	â	Absent	â	Biggest visible gap
Menus / settings UI	Present	â	Absent
Measurement	Min/max/Vpp/mean/RMS	â	Probable	â	Computed, not displayed
Frequency / duty / cursors	Unverified	â	Absent
Calibration	Probable, persisted	â	Math implemented, no storage	â	Unfinished
Controls	Range switch	â	Present	â	Decoded on ADC ch6 (all 3 verified)
Gain switch	Present	â	Decoded on ADC ch5 (9/9 combos verified)	â	Display scaling, not counts
Coupling AC/DC/GND	Present (analogue)	â	Unreadable	â	ch7 never moves	â	Hardware limitation

Interfaces
USB â Absent â power only (0 string hits) â Absent â Same

UART console
115200 (verified BRR=0x271) â Text + binary protocol v1

PC protocol
None known â 10-byte commands/replies, 520-byte frames, ~20 fps â Ours only

SWD diagnostics
None â status_* counters + live_check.py â Ours only

Reliability
Watchdog â v6 patch ~8 ms (RLR=4, PR=/64) â 10 s, correct handshake â Stock is hazardous

Bounded waits
Unknown â Every wait deadline-bounded

Verification
Host tests â None â 21 suites, make check, make guard â Ours only

2. Differences, and why

Stock has, ours lacks â all unfinished, none hardware-limited

- â¢ Menu system and on-screen settings.
- â¢ Displayed measurements (Vpp, mean, frequencyâƒ).
- â¢ Calibration data persisted in flash and applied per range.
- â¢ Timebase and trigger-level control from the device.
- â¢ A complete, product-tested user flow.

Ours has, stock lacks

- â¢ Documented binary PC protocol with CRC and ~20 fps streaming.
- â¢ Host-side regression tests, documentation/artefact checker, safety guard.
- â¢ SWD liveness counters and a scripted, evidence-based debug workflow.
- â¢ A correct 10 s watchdog (stock's v6 patch is ~8 ms â arguably a defect).
- â¢ Per-range sample-time policy and exact 64 MHz timebase maths.
- â¢ Column-mean rendering with a scale label derived from the switches.

Hardware-limited (neither firmware can change it)

- â¢ AC/DC/GND coupling is analogue-only: no ADC channel or GPIO observes it.
- â¢ The ADC injected group completes on this clone but returns zero data (JEOC set, JDR=0), so the vendor's injected trick is unusable here.

Intentional differences

Small testable architecture; PC protocol; trigger level auto-tracking instead of a knob; SysTick deliberately does not feed the watchdog, so a hung loop resets.

3. Why 60,584 B versus 18,132 B

Stock | Ours

Flash used
59,411 non-0xFF bytes (93.6 % of 62 pages) â 18,132 B (18 pages)

Layout
60 of 62 pages touched; largest erased run 2,904 B â .isr_vector 236; .text 15,532; .rodata 2,228; .data 100

Functions
not enumerable (stripped) â 143

Hypotheses for the ~42 KiB gap (ranked, not proven)

- â¢ Unimplemented features â menus, measurement displays, calibration storage and their strings/tables simply do not exist in our image. Capability gap, not bloat.
- â¢ Link-time dead-code elimination â we build with -ffunction-sections -fdata-sections -Os -Wl,--gc-sections; vendor images of this era are usually linked whole, retaining unused drivers and library routines.
- â¢ Data tables â 732 versus 258 strings, plus UI fonts/icons/menu text; ours is a single 5ˆ 7 font.
- â¢ Toolchain / libc â stock is almost certainly Keil/armcc with its own runtime and float formatting; we link newlib with integer-only formatting. Float printf alone can cost several KiB.
- â¢ Vendor extras â ATSHA204A one-wire driver, settings journaling, board-revision conditionals that are dead on this unit.
- â¢ Alignment/padding â scattered erased runs (largest 2,904 B) consistent with aligned sections and scatter-file padding.

Verified: byte counts, page usage, section sizes, function count, string counts. Inferred: the attribution between them â a definitive answer needs a vendor map file, which does not exist.

4. Where we stand

Our firmware is a working scope core with unusually strong verification, not yet a product: acquisition, triggering, timebase maths, display and the PC link are real and verified on hardware; range and gain decode across all nine combinations; the codebase is 3.3ˆ smaller than stock and leaves ~45 KiB of flash and ~13 KiB of RAM for the missing product layer.

Biggest remaining gaps, in order

- â¢ On-screen measurements and menus â the visible difference a user notices first.
- â¢ Calibration persistence â without it, volts/division is a computed label, not a calibrated instrument.
- â¢ Device-side timebase and trigger controls (protocol works; the panel doesn’t).
- â¢ Trigger mode selection â AUTO/NORM/SINGLE logic exists but is not wired.
- â¢ Coupling indication â permanently hardware-only; at best label it "set by switch".

Everything above is unfinished work in our tree, not a hardware ceiling â except coupling, which is genuinely outside the MCU’s reach on this board.